NOV. 7, 2026 · KENT STATE UNIVERSITY

Agentic workflows for robotics

Build with Codex and Claude Code

1–3 p.m. EST · MSB room 228

Advanced Telerobotics Research Laboratory
Department of Computer Science, Kent State University

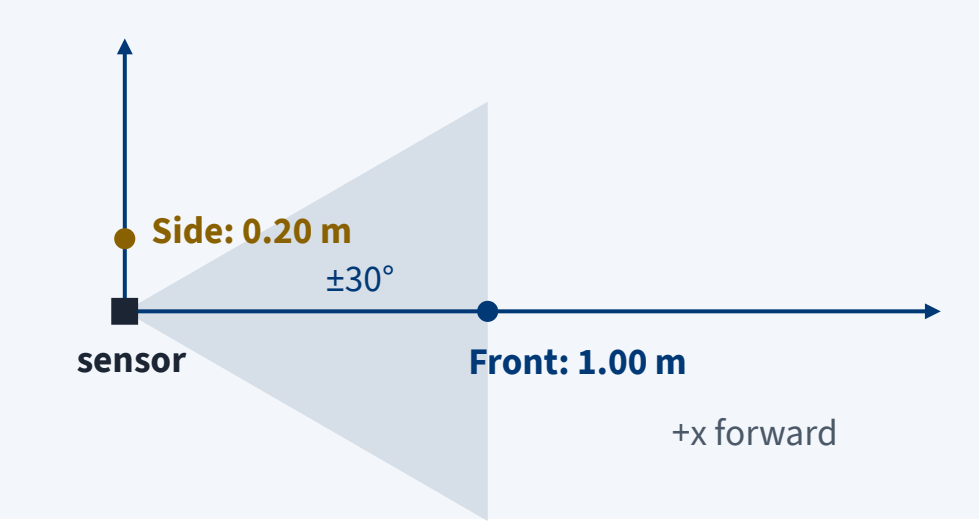


ADVANCED TELEROBOTICS RESEARCH

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Which distance answers the question?

- Side wall: 0.20 m away
- Forward point: 1.00 m away
- Both points have z = 0.20 m
- Predict the forward-sector result.



Authored fixture · top view

ROS 2 in three minutes

Driver node → **Typed topic** → **Your node**

Unitree ROS 2 driver or synthetic source

PointCloud2 metadata + bytes

Decode, transform and classify

Inspect: frame · timestamp · fields · stride · QoS

02 · SPECIFY AN OBSERVABLE OUTCOME

Write the acceptance contract

- Frame: x forward, meters
- Sector: ±30°; z: 0.05–0.50 m
- Metric: minimum hypot(x, y)
- Blocked: distance ≤ 0.50 m
- Age: 0–0.25 s
- Minimum points: 1
- Missing transform or data: unknown

unknown

is an explicit result.

It must never silently mean “clear.”

Give context that earns its place

Goal → **Interface** → **Checks**

One observable result

Frame, schema and time policy

Reproducer and acceptance cases

Keep rules in AGENTS.md; load the relevant skill.

Assign roles, then exchange evidence

Implementer

- Read the contract and failing case
- Propose one bounded change
- Show the diff and executed checks
- Explain what remains unverified

Reviewer

- Read the contract and code
- Trace one input through the diff
- Request a concrete reproducer
- Challenge assumptions and limits

Spend tokens until you can explain the change yourself.

Adapted from Theo Browne · explanation example



03 · THE OFFLINE REPRODUCER

Reproduce before editing

- Run the starter unchanged.
- Pick one failing fixture.
- Write expected and observed output.
- Ask for an explanation before a fix.

EXPECTED

clear · 1.00 m

STARTER

blocked · 0.20 m

Change which points count

Decode and TF	Validate data	Filter geometry	Reduce and classify
Metadata → XYZ Working frame	Clock, age finite values	Height band forward sector	Planar minimum unknown policy

Test the narrow change against the original contract.



Explain the result back

Ask the agent

- Trace the side point through the code.
- Why does the front point survive?
- What happens at exactly 30°?
- What happens after messages stop?

Verify independently

- Reference: 12 fixture checks
- Replay: reference node, synthetic bag
- Inspect the reported state
- Keep the true frontal-obstacle case

03 · EXECUTED ON NOV. 3, 2026

Three different checks passed

FIXTURE

12/12

Offline reference contract checks

ROS REPLAY

9/9

Synthetic bag phase and fault checks

PACKAGE

6/6

Collected pytest contract tests

Parallel work needs an interface

Perception	Replay	Review
Own the decoder and reducer	Own repeatable inputs and checks	Own the contract and counterexample

One integration owner · explicit ROS domains · one command path

Match claims to evidence

Fixture	Replay	Simulation	Physical trial
Reducer behavior	That stream, QoS and clock	Configured virtual robot	Prepared real setup

This workshop: synthetic replay + headless simulation

Build the whole application

LaserScan → **Clearance** → **State machine**

Native TurtleBot3 simulator sensor

Gap from the robot envelope

Freshness + limits Deterministic code

Command guard → **Gazebo base** → **Odometry + log**

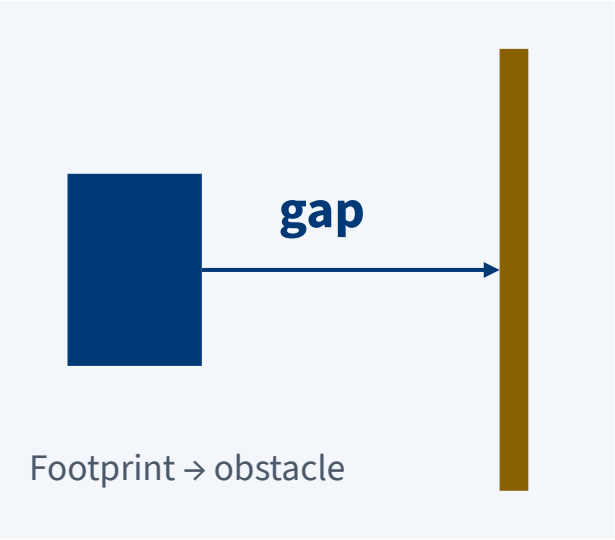
Independent receive timeout

Guarded velocity input only

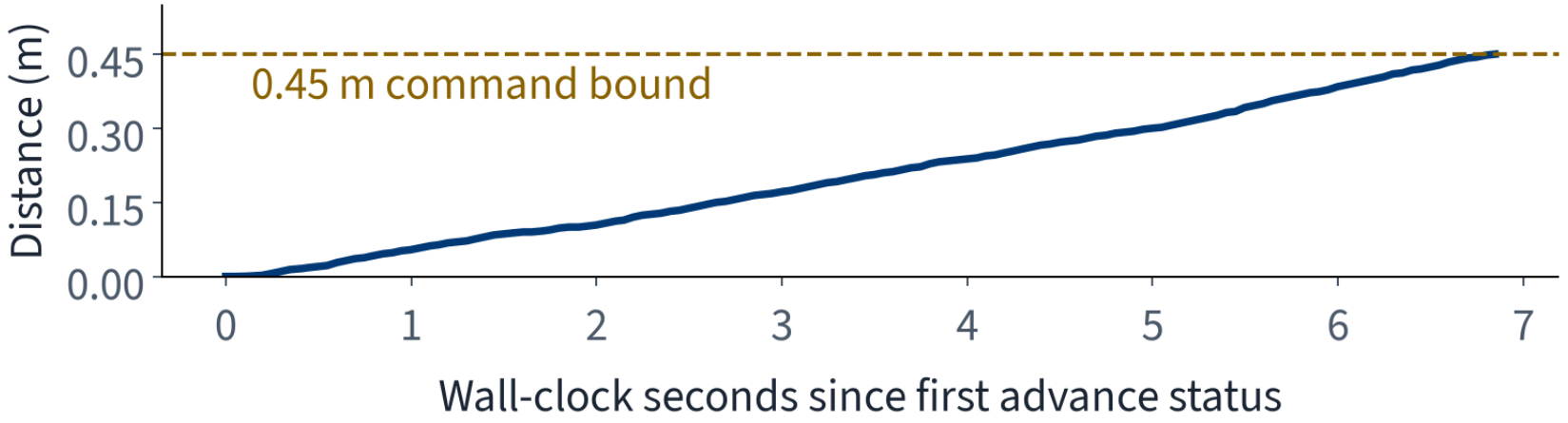
Observed response and stop reason

Use the robot envelope

- Diagnostic: radial distance from origin
- Capstone: gap ahead of the footprint
- Front extent: 0.11 m
- Swept corridor half-width: 0.15 m
- Stop gap: 0.30 m



Make the simulated path observable



SOURCE // saved Gazebo events · Nov. 3, 2026

A missing callback must stop motion

Sensing lost → **Periodic check** → **Zero command**

Drop / invalid / missing transform

Controller detects freshness failure

Fault remains latched until reset + start

Controller stalls → independent guard times out → zero

05 · EXECUTED ON NOV. 3, 2026

The simulation left a run record

SCENARIOS

10/10

Checks passed in one fresh world

DISTANCE

0.451

Meters observed for a 0.45 m bound

WATCHDOG

0.30 s

Configured command reception timeout

Leave an understood, repeatable change

Include in the handoff

- Revision and exact command
- Contract and relevant configuration
- Scenario outcomes and run artifacts
- Limits and the next bounded task

Explain in your own words

- Which behavior changed?
- Why is the new geometry correct?
- Which failure paths stop the robot?
- What has not been tested?

Choose your next bounded task



Find one reproducer

Write the contract

Keep the diff small

Run real checks

Explain it, hand it off

What input and observable result define your next task?

Questions?

From a prompt to evidence

Read · specify · change · verify · explain

Hands-on lab at 2 p.m. here
Mathematical Sciences Building
Room 228 · Kent State University

FIND THE LAB

WEB	atr.cs.kent.edu
EMAIL	atrlab.kent@gmail.com
X	@atrlab_kent
GITHUB	github.com/ATR-Lab

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ONE-HOUR HANDS-ON LAB · 2–3 P.M.

Make the workflow your own

01	Check and reproduce	T+00
02	Repair and explain	T+10
03	Replay through ROS 2	T+30
04	Run the simulation	T+38
05	Hand off, then Q&A	T+48

Programming experience

Robotics optional

Python 3.9+ and Docker

Existing agent login

One script per step

LAB A · STEPS 00–02 ON YOUR LAPTOP

Checkpoint 1: predict, then repair

- Check the machine: 00-check.sh
- Watch the starter fail: 01-reproduce.sh
- Ask the agent to explain, then fix
- Rerun 02-repair.sh until 12/12
- Keep check.py unchanged
- Explain one new boundary prediction

SAVE THE EVIDENCE

Input
Expected result
Observed result
Command
Limits

Checkpoint 2: replay and simulate

- Replay the bag: 03-replay.sh
- Run the simulation: 04-simulate.sh
- Read each stop reason
- Stuck? Rerun with doctor
- Draft the handoff: 05-handoff.sh
- Questions from about 2:50

SAVE THE EVIDENCE

Input
Expected result
Observed result
Command
Limits



SOURCES AND TAKE-HOME MATERIAL

Sources and reproducible materials

[1] Theo Browne: explanation/refinement example, ~20:34. youtu.be/q1D90-uGvBg?t=1234

[2] ROS 2 Jazzy PointCloud2 schema. github.com/ros2/common_interfaces/blob/jazzy/sensor_msgs/msg/PointCloud2.msg

[3] Unitree UniLiDAR SDK and ROS 2 driver. github.com/unitreerobotics/unilidar_sdk2

[4] ROBOTIS TurtleBot3 simulation manual. emanual.robotis.com/docs/en/platform/turtlebot3/simulation/

[5] Workshop: docs/start-here.md, lab-commands.md and participant-workbook.md

[6] Executed evidence: docs/verification.md and evidence/